

COTTESMORE, United Kingdom

WMO: 034530

Lat: 52.736N

Lon: 0.649W

Elev: 141

StdP: 99.65

Time Zone: 0.00 (EUW)

Period: 94-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.3	-2.9	-6.4	2.2	-2.0	-5.2	2.5	-1.0	15.4	9.0	14.0	8.9	3.1	270	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.1	17.7	23.9	17.0	22.1	16.2	18.7	23.5	17.8	22.2	17.0	20.9	5.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.4	20.4	16.2	11.7	19.2	15.4	11.1	18.4	53.5	23.5	50.7	22.4	48.1	20.9	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.7	10.4	DB	-6.9	29.4	1.9	1.4	-8.3	30.4	-9.4	31.2	-10.5	32.0	-11.9	33.0	(4)
(5)			WB	-7.2	20.3	1.8	1.2	-8.5	21.1	-9.6	21.8	-10.6	22.4	-11.9	23.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	3.9	4.3	5.9	8.5	11.2	14.2	16.3	16.3	13.9	10.6	6.7	3.6
	DBStd	5.50	3.19	3.25	3.02	3.06	2.99	2.81	2.63	2.60	2.48	2.93	3.21	3.71
	HDD10.0	901	191	161	129	63	21	1	0	0	2	28	106	198
	HDD18.3	3205	449	393	384	296	222	128	75	76	134	242	351	456
	CDD10.0	771	1	1	3	18	58	127	196	196	119	45	6	1
	CDD18.3	32	0	0	0	0	0	4	13	13	2	1	0	0
	CDH23.3	239	0	0	0	1	3	36	94	91	12	3	0	0
	CDH26.7	34	0	0	0	0	0	4	13	17	1	0	0	0
Wind	WSAvg	5.3	5.8	5.8	5.8	5.3	5.3	5.0	4.9	4.8	5.0	5.4	5.3	5.2
Precipitation	PrecAvg	628	52	44	37	45	45	50	58	59	55	62	60	52
	PrecMax	797	87	82	76	132	108	131	108	153	120	116	105	104
	PrecMin	449	13	10	5	3	8	8	12	7	8	24	31	19
	PrecStd	109	21	19	20	34	25	33	30	35	33	26	22	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.9	15.3	20.4	23.1	27.1	28.5	28.8	25.3	20.0	14.9	12.9
		MCWB	10.5	10.6	10.5	13.4	15.3	18.7	18.5	18.4	16.7	15.3	13.3	11.5
	2%	DB	11.0	11.4	13.6	17.3	20.7	23.9	26.0	26.0	22.0	17.2	13.6	11.6
		MCWB	9.5	9.6	9.9	11.8	14.8	16.7	17.6	17.8	16.3	14.2	12.1	10.4
	5%	DB	9.9	10.2	12.1	15.4	18.5	21.6	23.7	23.3	19.8	15.9	12.4	10.2
		MCWB	8.6	8.4	9.2	10.7	13.5	15.8	17.1	17.0	15.0	13.6	10.9	9.1
	10%	DB	8.7	9.1	10.9	13.7	16.6	19.6	21.7	21.3	18.3	14.8	11.2	8.9
		MCWB	7.5	7.6	8.4	9.8	12.3	14.6	16.0	16.0	14.1	12.8	9.9	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.5	12.1	14.1	16.9	19.6	19.7	20.0	18.2	16.3	13.6	11.7
		MCDB	11.8	12.4	13.8	18.7	20.9	25.2	25.0	24.4	22.5	19.3	14.5	12.5
	2%	WB	9.7	10.0	10.7	12.6	15.3	17.5	18.6	18.7	17.0	15.1	12.4	10.6
		MCDB	10.7	11.2	12.6	16.2	19.7	22.1	23.7	23.2	20.5	16.7	13.3	11.5
	5%	WB	8.8	8.7	9.7	11.4	14.0	16.3	17.7	17.8	16.0	14.1	11.2	9.2
		MCDB	9.7	9.8	11.5	14.6	17.7	20.2	22.2	22.0	18.6	15.5	12.0	10.0
	10%	WB	7.6	7.7	8.8	10.2	12.8	15.3	16.8	16.9	15.0	13.2	10.1	8.1
		MCDB	8.6	8.9	10.5	13.1	15.9	18.8	20.8	20.5	17.4	14.6	11.0	8.9
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	6.8	8.0	8.3	8.5	8.8	8.6	7.6	6.4	5.3	4.6
		MCDBR	5.8	6.2	8.3	10.4	11.1	11.7	11.9	11.6	9.6	6.9	5.6	5.5
		MCWBR	5.5	5.1	5.4	5.9	6.3	5.9	5.2	4.9	4.8	4.7	4.9	5.3
	5% WB	MCDBR	5.7	6.0	7.3	9.5	10.3	10.3	10.4	10.0	8.5	6.5	5.5	5.4
		MCWBR	5.6	5.3	5.3	5.9	6.3	5.9	5.2	4.8	5.1	5.1	5.1	5.3
Clear-Sky Solar Irradiance	taub	0.320	0.335	0.367	0.393	0.397	0.399	0.399	0.387	0.367	0.348	0.322	0.309	
	taud	2.364	2.393	2.297	2.233	2.262	2.278	2.285	2.345	2.407	2.442	2.416	2.341	
	Ebn at Noon	664	773	814	837	853	853	845	836	810	747	659	602	
	Edh at Noon	60	77	103	124	127	126	123	110	92	73	56	51	
All-Sky Solar Radiation	RadAvg	0.73	1.43	2.50	3.83	4.75	5.04	4.87	4.01	3.00	1.74	0.91	0.60	
	RadStd	0.07	0.17	0.30	0.43	0.31	0.49	0.52	0.28	0.27	0.15	0.10	0.06	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page